

Beliefs as Steerable Attractors:

The Geometry and Causal Structure of Evidence Integration
in Model-Free Reinforcement-Learning Agents

Abstract

Recurrent reinforcement-learning (RL) agents acting under partial observation must maintain a *belief* about the unobserved state of their environment. It is by now routine to *decode* such beliefs from an agent’s hidden state, but decodability alone says little about what the belief *is* — a static register that stores a value, or a dynamical process — or whether it *causes* the agent’s behavior. We study belief in model-free RL agents as a dynamical, causal object across three partially observed decision tasks that differ only in the *structure of the evidence* they present: a categorical cue, a stream of accumulating left/right evidence, and noisy naturalistic terrain. We find, using one analysis pipeline, that (i) the agent’s belief occupies a low-dimensional manifold in recurrent-state space whose geometry is dictated by the task’s evidence model — a set of discrete attractors for categorical evidence, and a one-dimensional line attractor for evidence accumulation (net evidence linearly decodable at $R^2 = 1.00$); (ii) this belief is an integrating attractor, not a register: it forms gradually, persists across gaps in observation with no external cue, and revises in a lawful direction when evidence is forced off-policy; and (iii) behavior is a graded, monotone function of position on the belief manifold — clamping the hidden state at points along the manifold traces a smooth psychometric curve that gives *full* control of the agent’s choice — yet the *same* intervention is behaviorally inert in a control task where the belief is represented and even persistently implantable but not decision-relevant. This double dissociation shows that causal load-bearing is a property of the task’s decision structure, not of representational quality. Along the way we show that the choice of belief metric matters: a categorical (Hellinger) embedding is faithful for unordered latents but folds an ordered scalar latent into a misleading horseshoe, which a magnitude-respecting metric repairs. Our accumulation task instantiates, in a reward-trained agent with full observability and exact causal control, the line-attractor-and-perturbation paradigm long studied in decision neuroscience.

1 Introduction

An agent acting under partial observation cannot see the state that determines the consequences of its actions. To act well it must maintain a *belief*: a summary of its history sufficient for choosing actions (Kaelbling et al., 1998). In deep RL this belief is not designed but learned — typically it lives, implicitly, in the hidden state of a recurrent policy (Lambrechts et al., 2022; Mikulik et al., 2020). A growing body of interpretability work shows that such beliefs can be *read out*: linear probes recover the latent variables an agent should track, and recent work even recovers the predicted *geometry* of the optimal belief in sequence models (Shai et al., 2024; Li et al., 2023). Decoding a variable, however, leaves the scientifically central questions untouched. It does not tell us what *kind* of object the belief is — a value latched into a register, or a process unfolding in the agent’s dynamics — and, notoriously, it does not tell us whether the decoded variable actually *causes* the agent’s behavior or is merely correlated with it (Ravichander et al., 2021).

We take these two questions — *what form?* and *does it cause?* — as the subject of the paper, and study them in model-free RL agents, where belief is a by-product of reward maximization rather

than an explicitly trained objective. Our stance is that a belief is best understood as a *dynamical, causal* object, and that the way to characterize it is to (i) look at the *geometry* traced out by the agent’s recurrent states, (ii) analyze the *dynamics* that move states through that geometry, and (iii) *intervene* on those states and measure the effect on behavior. This viewpoint is standard in the systems-neuroscience study of decision circuits, where trained recurrent networks and neural populations alike are reverse-engineered into low-dimensional dynamical pictures — fixed points, line attractors, and their perturbations (Mante et al., 2013; Sussillo & Barak, 2013; Wong & Wang, 2006). We bring it to bear on model-free RL agents, where we can do what neuroscience cannot: observe the belief with ground truth and steer it with surgical precision.

The key design choice is to hold the analysis fixed while varying the *structure of the evidence* a task provides (Figure 1). In a *categorical-cue T-maze*, a single symbolic cue seen early must be carried to a decision made later — the belief is a discrete class. In an *accumulation T-maze* — the perceptual decision-making paradigm of Brunton et al. (2013) and Pinto et al. (2018), rendered as navigation — a stream of left/right evidence must be integrated and the majority side chosen; the belief is a *scalar* accumulator. In a *naturalistic navigation* task, the belief (which terrain regime the agent is in) must be inferred from noisy, distributed observations. Because the tasks differ only in evidence structure, differences in the recovered belief object attach to that structure, and commonalities attest to the method’s generality.

Contributions. We make three claims about belief in model-free RL agents, each supported by the same pair of tools across all three tasks.

Claim 1 (Geometry follows evidence structure). *The belief occupies a low-dimensional manifold in recurrent-state space whose geometry is dictated by the task’s evidence model: discrete attractor points for categorical evidence and a one-dimensional line attractor for scalar accumulation. One task-agnostic construction recovers the correct object in each case.*

Claim 2 (Belief is an integrating attractor, not a register). *The belief forms gradually as evidence arrives, persists across an observation gap with no external cue to sustain it, and revises in a lawful direction when fresh evidence is forced off-policy; the recurrent dynamics show marginally stable slow manifolds consistent with integration rather than a latched value.*

Claim 3 (Behavior is a graded causal function of manifold position, and this is task-contingent). *Clamping the hidden state at fractional positions along the belief manifold produces a smooth psychometric curve over choice with full control at the endpoints. Yet in a control task where the same latent is represented — and can even be persistently implanted — but is not decision-relevant, the identical intervention is behaviorally inert: a double dissociation showing that causal load-bearing is a property of the task’s decision structure, not of decodability.*

Why it matters. For interpretability, Claim 3 upgrades “we can decode X ” to “ X is a causal, continuously manipulable variable,” and gives a graded test — the manifold-clamp psychometric — for whether a probed feature is load-bearing, sharper than an all-or-nothing patch. For the science of learned agents, viewing belief as attractor dynamics explains why memory persists across observation gaps and predicts how it fails. And because our accumulation task reproduces, in a reward-trained agent, the line-attractor mechanism and perturbation logic foundational in decision neuroscience (Mante et al., 2013; Hanks et al., 2015), it offers an in-silico model organism where those experiments can be run with full observability and exact causal control.

2 Related work

Belief representations in RL and sequence models. That recurrent POMDP agents encode belief-like sufficient statistics is established theoretically and empirically (Kaelbling et al., 1998; Lambrechts et al., 2022), and meta-RL agents can approximate Bayes-optimal updates (Mikulik et al., 2020). Recent interpretability decodes such latents and even recovers the *geometry* of the optimal belief predicted by computational mechanics in transformers (Shai et al., 2024), echoing emergent world-state representations in game-playing models (Li et al., 2023). We depart in two ways: we study *model-free* agents where belief is incidental to reward, and we go past decoding to *causal*, *graded* intervention and to *dynamical* characterization.

Dynamics of decision-making in trained RNNs and brains. A long tradition reverse-engineers recurrent computation into low-dimensional dynamics — fixed points and line attractors underlying integration and choice (Sussillo & Barak, 2013; Mante et al., 2013; Wong & Wang, 2006), with evidence accumulation as the canonical case (Gold & Shadlen, 2007; Brunton et al., 2013). Causal perturbations of the accumulator in cortex sharpened this picture (Hanks et al., 2015), and belief-state accounts connect it to reward learning (Starkweather et al., 2017). Our accumulation environment is a deliberate RL analogue of the accumulating-towers task (Pinto et al., 2018); our contribution is to show the same dynamical object arising in a *model-free*, *reward-trained* agent and to steer it along its manifold.

Activation-level interpretation and steering. Causal mediation via activation patching and representation-level steering vectors are common tools for locating and manipulating features (Meng et al., 2022; Turner et al., 2023; Zou et al., 2023), and probing studies caution that decodability need not imply causal use (Ravichander et al., 2021). Rather than restate that caution, we operationalize its resolution: a manifold-clamp psychometric that measures *how much* of the behavior a belief controls, and a matched-representation control task that isolates *when* a represented belief is load-bearing.

3 Setup and definitions

Partially observed decision tasks. Each task is a POMDP in which a hidden latent z — the categorical cue, the sign/magnitude of accumulated evidence, or the terrain regime — determines which terminal choice is rewarded. The agent receives egocentric symbolic observations o_t and emits actions a_t ; a recurrent policy $\pi_\theta(a_t | h_t)$ carries a hidden state $h_t = f_\theta(h_{t-1}, o_t)$ (a GRU). All three tasks share an observation and action interface, so a single architecture and analysis pipeline apply unchanged. We train model-free recurrent agents with PPO (Schulman et al., 2017) to high success and freeze them for analysis; the accumulation agent reaches 100% success (optimal-length trajectories), and the cue agents reach 100%.

Belief. Operationally, the belief at time t is a distribution $b_t = g(h_t)$ over the task-relevant latent, where g is a *linear* read-out (a multinomial logistic probe) fit on held-out episodes from the frozen hidden states. Linearity keeps the belief an affine function of h_t , so geometry in belief space corresponds to geometry in hidden space and interventions can be expressed in either. We stress that b_t is a *measurement* of the agent’s state, not a claim that the agent computes it explicitly; the causal experiments (§8) license statements about what the belief *does*.

4 Method

One triplet of tools, applied identically to every task.

(1) Belief-space geometry. From many frozen rollouts we collect hidden states $\{h_t\}$ and probe beliefs $\{b_t\}$. To visualize the shape of the set of beliefs we embed them with a divergence-preserving projection: map each belief through the Hellinger embedding $b \mapsto \sqrt{b}$ and take leading principal components (equivalently, information-geometric MDS on the Bhattacharyya divergence). Confident beliefs land at simplex vertices and ambiguous beliefs in the interior, so the number of distinguishable states and how the agent moves between them become directly visible and comparable across tasks. For *ordered* latents we additionally use a magnitude-respecting embedding (§6.1).

(2) Attractor analysis. To ask whether the belief is a dynamical attractor we analyze the frozen recurrent map $h \mapsto f_\theta(h, o)$ under a fixed post-evidence observation, locating slow points by minimizing the update magnitude $\|f_\theta(h, o) - h\|^2$ and reading local stability from the Jacobian spectrum. Marginally stable directions (eigenvalues near the unit circle) signal integration/memory — a continuum of near-fixed states along which evidence pushes the belief and where it rests.

(3) Manifold-clamp steering. To test causal load-bearing in a graded way we intervene on the hidden state. Given the belief manifold’s local axis $\hat{u}(x)$ at task-phase x (the direction between class-conditional means, made position-local because the code can rotate along a trajectory), we clamp the component of h_t along $\hat{u}$ to a target ρ_f interpolating between the endpoints at fraction $f \in [0, 1]$:

$$h_t \leftarrow h_t + (\rho_f - \langle h_t, \hat{u} \rangle) \hat{u},$$

sustained through the decision phase. Sweeping f and reading out the terminal choice yields a *psychometric curve*: behavior as a function of implanted belief. A smooth, monotone curve with full endpoint control means the belief is continuously steerable and causally load-bearing; a flat curve means it is not.

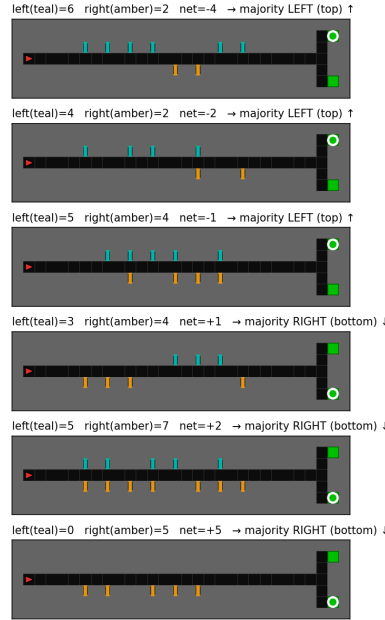
5 Tasks

The three environments span the axis of evidence structure while sharing interface, architecture, and analysis.

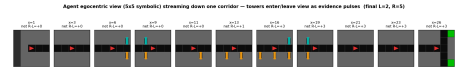
Categorical cue (memory; discrete belief). A T-maze in which a single cue, visible only briefly near the start, specifies the rewarded branch/door; the belief is one of K classes. We use $K \in \{2, 3, 4\}$ cue sets. A cue carries two attributes (a shape indicating the branch and a colour indicating the door); the 2-cue set entangles them (two cues) while the 4-cue set factorizes them (all shape×colour combinations), which lets us test whether the agent’s belief factorizes. The cue leaves view long before the choice, so the task exercises memory (persistence) most starkly.

Evidence accumulation (towers; scalar belief). The neuroscience-canonical case as navigation (Figure 1): a stream of left/right “towers” must be integrated and the *majority* side chosen after the evidence leaves view. The belief is a scalar accumulator (net right–left).

Towers env — accumulation-of-evidence T-maze: a selection of maps
agent (red) runs east; teal = LEFT evidence (top wall), amber = RIGHT evidence (bottom wall); reach the majority-side goal (white ring)



(a) accumulation T-maze: a selection of maps



(b) egocentric evidence stream

Figure 1: The accumulation T-maze (towers). The agent runs east; a tower on the top wall is left evidence, on the bottom wall right evidence. After the evidence ends and leaves view, a T-junction requires choosing the majority side (white ring). (b) The egocentric 5×5 view: towers enter and leave as discrete evidence pulses while the running accumulator climbs. The three environments share this observation/action interface.

Naturalistic navigation (bridge_tunnel; noisy belief). A procedurally generated terrain-crossing task in which the belief — the terrain regime (rocky / lakes / balanced) that dictates the correct strategy — must be inferred from noisy, spatially distributed observations. It supplies the crucial *control*: two variants represent the same belief but differ in whether it is decision-relevant. In *fork_wall*, terrain determines which of two doors is correct (belief is load-bearing); in *commit*, the agent commits to a crossing strategy where the belief need not gate the choice.

6 Experiment 1: belief geometry follows evidence structure

We construct the belief-space embedding (§4) for each frozen agent (Figure 2). The geometries differ exactly as the evidence structure predicts, from one construction.

Categorical $\Rightarrow$ discrete attractors. For *memory*, the belief embeds as isolated vertices of a probability simplex, one per cue class: K classes occupy K vertices spanning $K-1$ dimensions (with a small residual). Confident beliefs are mutually equidistant, so the class *identity* — not a magnitude — is what the geometry encodes.

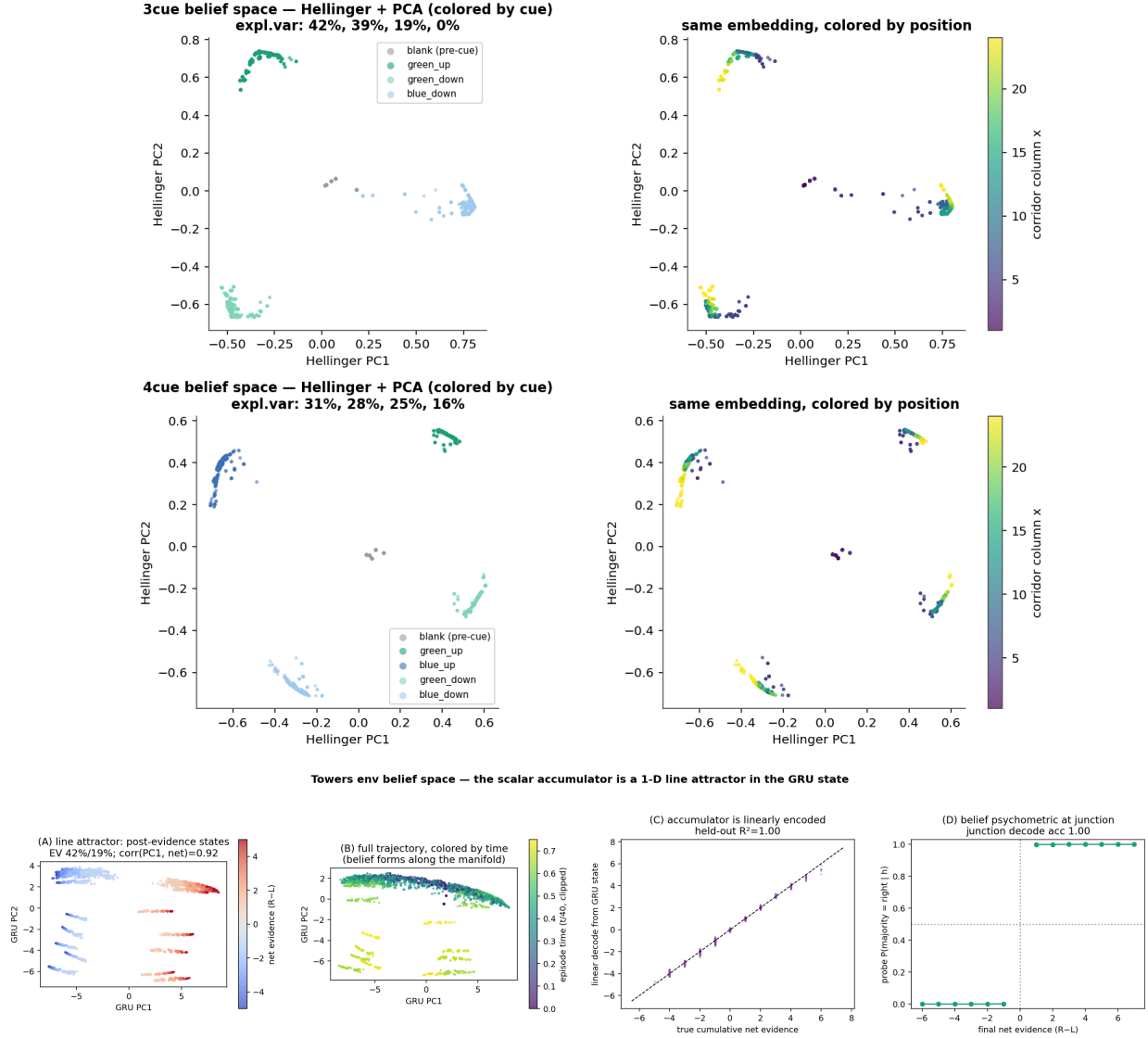


Figure 2: Belief geometry is dictated by evidence structure. *Top:* the memory categorical belief embeds as *discrete vertices* of a probability simplex (here the 3/4-cue sets); confident beliefs are mutually equidistant and the number of occupied vertices equals the number of cue classes. *Bottom:* the **towers** scalar belief is a *one-dimensional line attractor*: net evidence is linearly decoded from the GRU state at $R^2 = 1.00$ (C); post-evidence states form a single axis ordered by net evidence (A, correlation 0.92 with the leading PC); the belief forms along this axis over time (B); and the belief read-out is the sharp step of a near-perfect integrator (D).

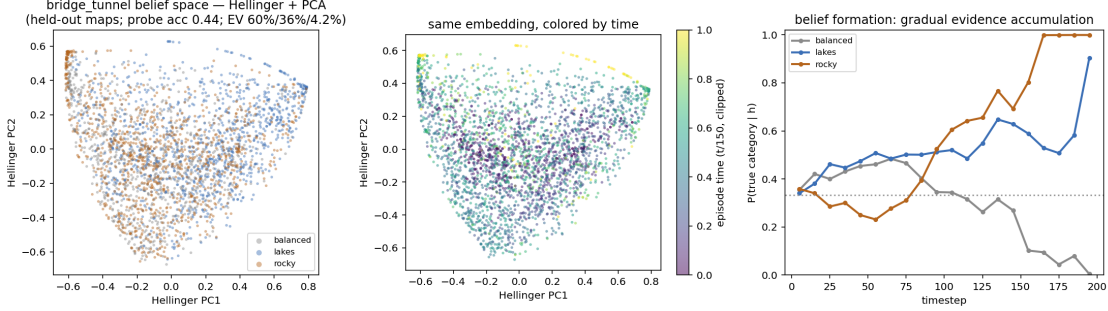


Figure 3: Naturalistic belief (bridge_tunnel). The map-category belief (rocky/lakes/balanced) inferred from noisy terrain, embedded with the same Hellinger construction and evaluated on held-out maps; belief forms gradually over the episode (right) rather than snapping to a vertex.

Scalar $\Rightarrow$ line attractor. For `towers`, the belief is a one-dimensional line. Net accumulated evidence is decoded from the GRU state by a single linear map at $R^2 = 1.00$; post-evidence states arrange along one dominant axis ordered by net evidence (leading principal component correlation 0.92); and early states start near the center and sweep outward as evidence arrives — integration, not a latched value. The attractor analysis (§7) confirms the axis is a slow manifold.

Naturalistic $\Rightarrow$ a graded, noisier manifold. For `bridge_tunnel`, the same construction yields a category belief that forms gradually from distributed terrain evidence, with the interior of the simplex genuinely occupied during inference (Figure 3). The belief is weaker in a plain agent (held-out category decode ≈ 0.48) and sharpened by an auxiliary belief head (≈ 0.76), but present in both.

6.1 The belief metric must match the latent’s structure

The Hellinger/Bhattacharyya embedding treats belief classes as *unordered*. This is faithful for the categorical cue — there is no magnitude to represent, and the equidistant simplex vertices are the honest picture. For the *ordered* scalar of `towers`, however, it is actively misleading: the Hellinger distance *saturates* at $\sqrt{2}$ as soon as two beliefs stop overlapping, so the maximally opposite beliefs (net +4 vs -4) sit at the same ceiling distance as moderately different ones, and a linear projection of this bounded, ordered metric folds into a *horseshoe* in which the extremes curl together (Figure 4A; projected distance 0.08 for the opposite extremes versus 0.15 for adjacent classes — an inversion of the truth). A casual reader would misread the folded extremes as similar beliefs.

Two remedies recover a faithful picture (Figure 4B,C): a magnitude-respecting distribution metric (Wasserstein-1, the L_1 distance of CDFs on the ordered support), whose classical MDS lays the beliefs on a straight line with the extremes maximally far ($d(+4, -4) = 7.76$ vs. $d(+4, +3) = 0.90$, all variance on one axis); or simply the scalar read-out itself, $\mathbb{E}[\text{net} \mid b]$ against belief confidence, which cannot fold. The methodological point generalizes: *match the belief metric to the latent’s structure* — categorical $\rightarrow$ Hellinger; ordinal/scalar $\rightarrow$ Wasserstein or the raw scalar axis.

Making the extreme beliefs far apart: don't Hellinger-embed a scalar latent — use a magnitude-respecting metric

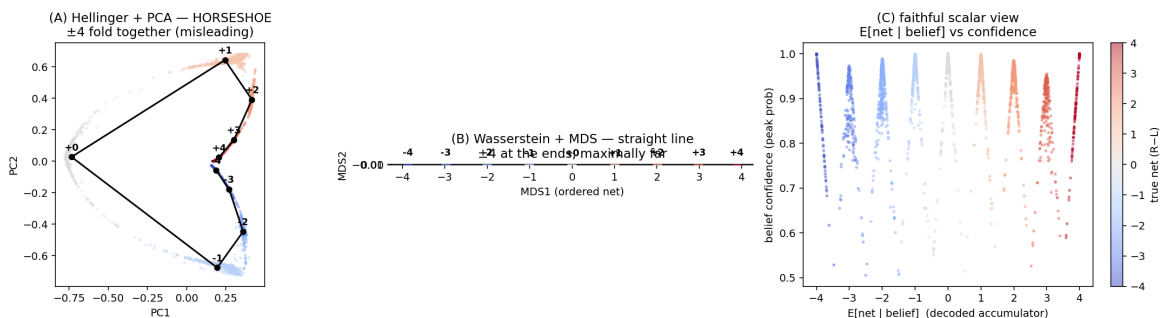


Figure 4: The metric encodes an assumption about the latent. (A) Hellinger+PCA of the ordered scalar belief folds into a horseshoe: the opposite extremes ± 4 project close together (misleading). (B) Wasserstein+MDS lays the same beliefs on a straight ordered line with the extremes maximally far. (C) The scalar read-out $\mathbb{E}[\text{net} | b]$ vs. confidence is faithful by construction.

7 Experiment 2: beliefs are integrating attractors, not registers

Formation and persistence. In all tasks the belief *forms* gradually — it moves along the manifold from ambiguity toward a vertex/endpoint as evidence arrives (Figure 2B for **towers**; the cue tasks snap to a vertex once the cue is seen and the naturalistic task ramps over many steps). Crucially the belief then *persists* across an observation gap with nothing sustaining it: in **towers** the accumulator is held stably through the run-out corridor after all towers leave view, and in **memory** the cue class is held from the cue room to the distant door. The attractor analysis of the frozen recurrent map finds no exact fixed points but hundreds of slow points with leading Jacobian eigenvalue magnitude ≈ 1.000 — approximately marginally stable slow manifolds (one per belief fiber), the dynamical signature of integration/memory rather than a latched register.

Lawful revision under forced evidence. We test integration causally by forcing the agent, off-policy, through evidence that contradicts its current belief and reading out the resulting belief (Figure 5). In **memory**, forcing the 2-cue agent down the wrong corridor *collapses its belief to the opposite attractor* — it comes to “believe” the other cue and its downstream door choice flips accordingly; the 4-cue agent instead *revises only the branch/direction factor* while preserving the colour factor, and the 3-cue agent revises in the direction consistent with its training likelihoods. The belief is thus not a frozen record of the cue but a state that integrates new evidence.

2cue: action-forced into the WRONG corridor — belief at the marker evidence (5-class probe)

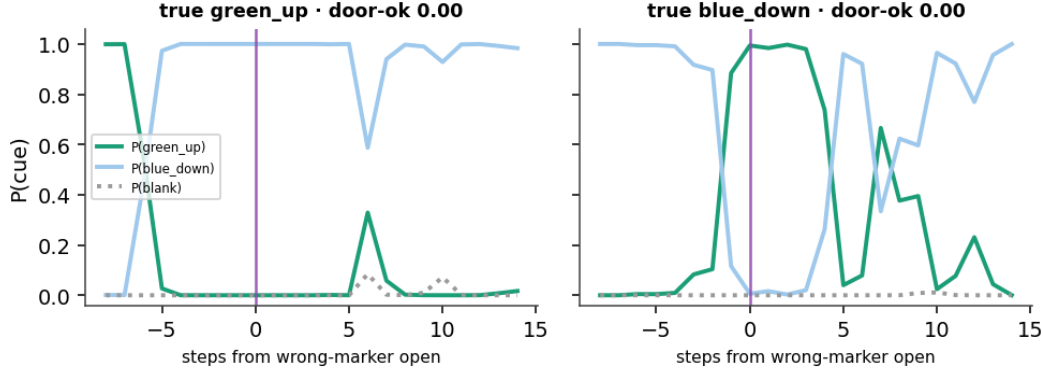


Figure 5: Beliefs integrate forced evidence (memory, 2-cue). Forcing the agent through the wrong corridor drives its decoded belief from one attractor to the other (aligned to the moment the marker is opened), and its downstream choice follows — the belief is updated by evidence, not merely stored.

8 Experiment 3: behavior is a graded causal function of manifold position

The central experiment clamps the hidden state at fractions f along the belief manifold at the decision phase and reads out the choice (§4).

towers: full graded control, evidence-independent. Clamping the accumulator belief traces a clean psychometric (Figure 6B): $P(\text{right door})$ moves $0.00 \rightarrow 0.03$ ($f=0.5$) $\rightarrow 0.97$ ($f=0.62$) $\rightarrow 1.00$, with *zero* timeouts — full control at both endpoints. The transition is steep (the sharp sigmoid of a near-perfect integrator). Splitting by the map’s true majority (Figure 6C), true-left and true-right maps follow the *same* curve: the implanted belief *overrides* the accumulated evidence, and the decision is a single path-independent function of the belief coordinate. The Hellinger belief arc (A) is shown for continuity with the other tasks; per §6.1 it is the saturated image of the line attractor.

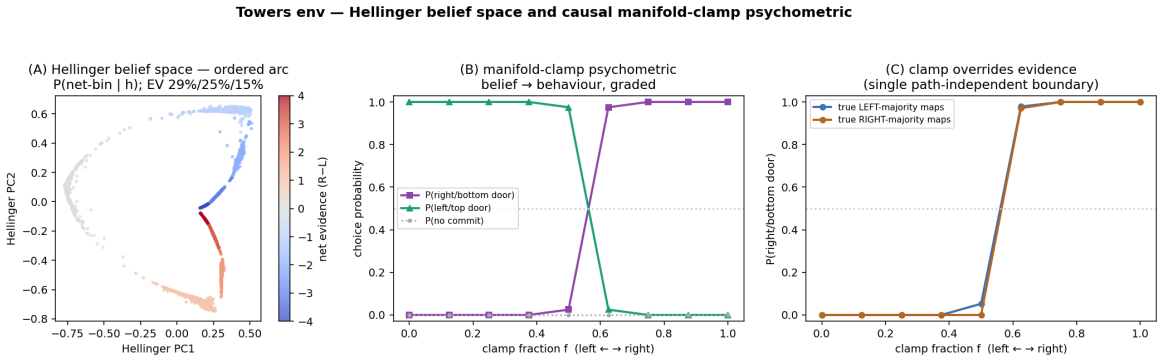


Figure 6: Manifold-clamp psychometric (towers). (A) Hellinger belief space (ordered arc). (B) Clamping the GRU belief along the manifold gives full graded control of the door with no timeouts. (C) True-left and true-right maps follow one curve: the clamp overrides the evidence (path-independent boundary).

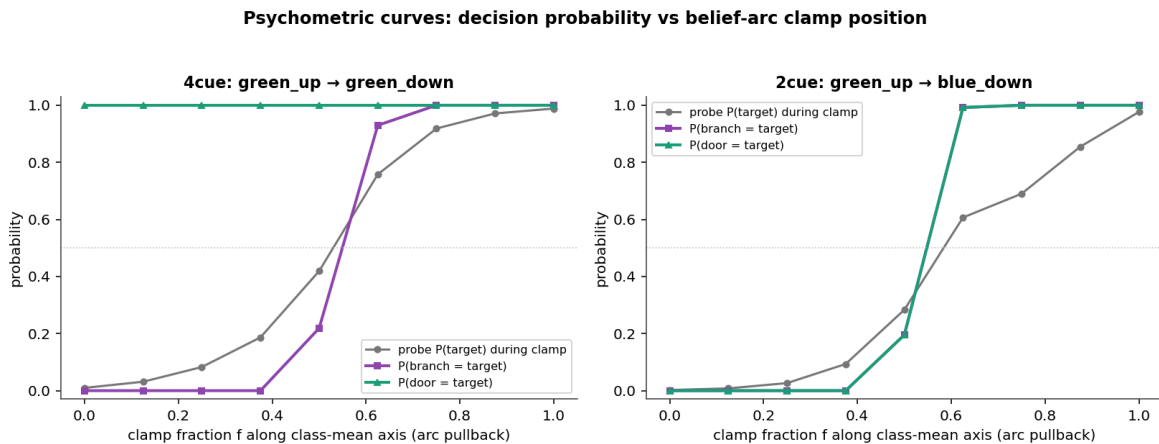


Figure 7: Manifold-clamp psychometric (memory). Probe belief (grey) moves smoothly with clamp position; behavior is a steep sigmoid. *Left*, 4-cue: clamping the shape axis flips the branch (purple) while the door (green) stays at 1.0 — the intervention respects the belief’s factorization. *Right*, 2-cue: branch and door are yoked and flip together.

memory: the psychometric respects the belief’s factorization. The same clamp along the cue-belief manifold (Figure 7) moves the probe belief smoothly and the behavior as a steep sigmoid. In the entangled 2-cue agent the branch and door choices are *yoked* — both flip together at the same f , because a single cue code carries both attributes. In the factorized 4-cue agent, clamping the shape/direction axis flips the branch smoothly while the door stays *invariant* at $P = 1.0$ across the entire edge, because the colour factor — and thus the door — is left untouched. The causal geometry of the intervention mirrors the representational geometry of the belief.

bridge_tunnel fork.wall: graded control in a noisy task. The same clamp on the naturalistic category belief gives a graded, monotone psychometric over the door choice (Figure 8), with full control at $f=1$ in both sweep directions. Sweeping from lakes toward rocky the choice crosses 50/50 at $f \approx 0.62$; sweeping from rocky toward lakes it crosses at $f \approx 0.38$ — *the same physical point* ($\approx 62\%$ toward rocky), a single path-independent boundary. The transition is broader than in **towers/memory**, consistent with ambient terrain evidence partially contradicting the implant at intermediate doses. Belief is load-bearing here.

bridge_tunnel commit: the belief is epiphenomenal — a double dissociation. In the **commit** variant the category belief is present and decodable (auxiliary agent held-out ≈ 0.76) and can even be *persistently implanted*: a full belief swap during an early window survives long after release, the decoded belief staying on the wrong category for 100^+ steps against contradicting terrain (Figure 9). Yet the agent’s behavior is *unchanged* — its commitment matrix and success rate are identical to baseline. The identical intervention that fully controls choice in **fork.wall** is behaviorally inert in **commit**. Representational quality, decodability, and even persistence do not make a belief causal; the task’s decision structure does.

Summary of the causal picture. Table 1 collects the three environments. The same manifold-clamp intervention gives three transfer functions on one axis — steep-full, factorized, graded-full, and flat — selected by the task’s decision structure, not by how well the belief can be read out.

BT fork_wall psychometric: door choice vs belief-arc clamp position

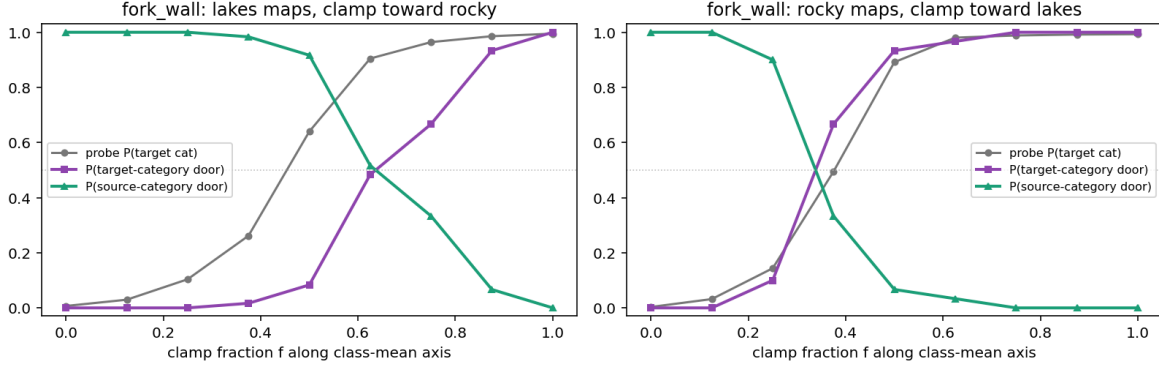


Figure 8: Manifold-clamp psychometric (bridge_tunnel fork_wall). Door choice varies smoothly with clamp position and is fully controlled at $f=1$ in both directions; the 50/50 boundary is the same physical point from either sweep ($\approx 62\%$ toward rocky) — path-independent.

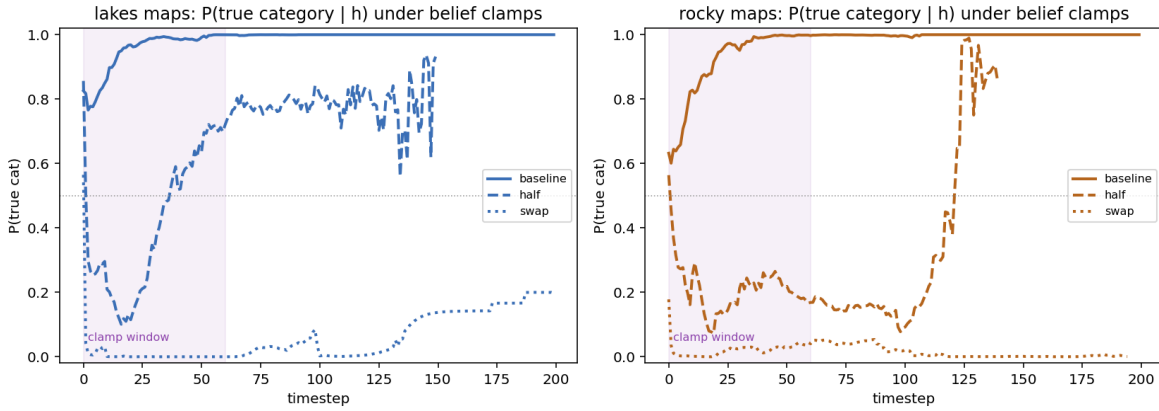


Figure 9: Persistent but epiphenomenal belief (bridge_tunnel commit). A full belief swap (dotted) during the clamp window sticks after release — the decoded belief remains on the wrong category for 100^+ steps — while a half-belief (dashed) re-corrects. Despite the persistent implant, the agent’s commitment behavior is unchanged from baseline (text): the belief is represented but not decision-relevant.

Environment	Latent	Belief geometry	Clamp $\rightarrow$ behavior	Belief causal?
memory (2-cue)	categorical (entangled)	simplex vertices	steep; branch+door yoked	yes
memory (4-cue)	categorical (factorized)	simplex vertices	steep; branch flips, door fixed	yes (factored)
towers	scalar accumulator	line attractor	steep, full, evidence-overriding	yes
bridge fork_wall	category (noisy)	graded simplex	graded, full, path-independent	yes
bridge commit	category (noisy)	graded simplex	flat (implant persists, inert)	no

Table 1: One intervention, four transfer functions, selected by decision structure. Decodability is high in all rows; causal load-bearing is not.

9 Discussion

Across three tasks that differ only in evidence structure, a model-free RL agent represents its task-relevant belief as a *dynamical attractor manifold* whose geometry mirrors the task’s evidence model, integrates evidence over time, persists without external support, and — where the task makes it decision-relevant — is a graded, continuously steerable cause of behavior. The manifold-clamp psychometric turns “is this feature causal?” into a quantitative dose–response curve, and the `commit` control shows why that matters: decodability, representational sharpness, and even implant persistence can all be present in a belief that does nothing. We also found that the *metric* used to visualize belief is a modeling choice with teeth — a categorical embedding folds an ordered latent into a misleading horseshoe — and gave a simple rule for matching metric to latent.

Scope and limitations. The agents are small recurrent policies on gridworld-scale POMDPs, chosen for the ground truth and causal access they afford; claims about “model-free RL agents” should be read as claims about this regime, not about frontier-scale agents. The belief is a linear measurement of the hidden state; non-linear structure it misses is a caveat we probe but cannot fully exclude. “Integration” and the neuroscience analogy are about consistency with the task’s evidence statistics up to the training distribution, not optimal Bayesian inference. Finally, the manifold-clamp axis is estimated from class-conditional means; we make it position-local to handle rotation of the code along a trajectory, but a mis-estimated axis would under-state causal control (a conservative bias).

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